

Lecture 7

Public Goods

Byeong-Hak Choe

bchoe@geneseo.edu

SUNY Geneseo

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Public Goods

Rivalry & Excludability

	Excludable	Non-Excludable
Rivalrous	Private Goods food, clothing, cars, personal electronics	Common Goods fish stocks, timber, coal
Non-Rivalrous	Club Goods cinemas, private parks, satellite tv	Public Goods air, national defense

Rival? Does one person's use reduce what remains available for others?

Excludable? Can access be restricted to those who pay—or can anyone consume the good even without paying?

Where Do Public Goods Show Up in Energy & Climate?

- **Basic R&D** in energy efficiency, storage, fusion, green hydrogen (knowledge spillovers).
- **Awareness campaigns** on climate & energy saving.
- **Protective infrastructure** (often *local* public goods):
 - Sea walls & surge barriers (coastal protection)
 - Riverbank & floodplain defenses
 - Urban heat mitigation (cool roofs, urban canopy)
- **Global mitigation** (e.g., CO₂ reduction benefits) vs. **local adaptation** (place-specific benefits).

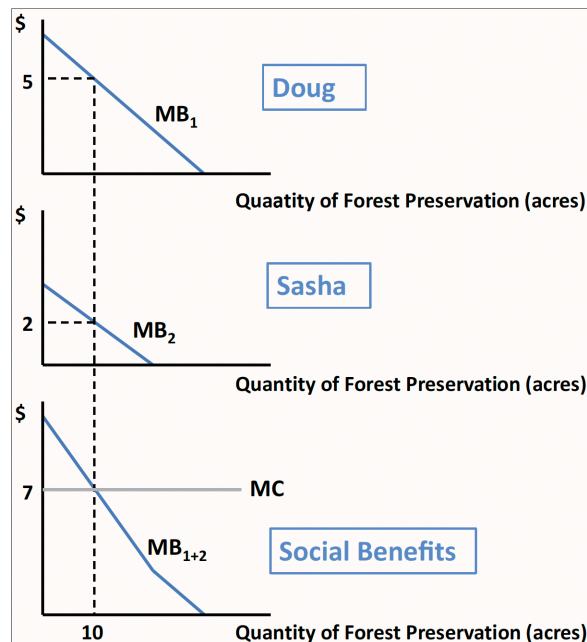
Market Failures - Why Markets Underprovide Public Goods

- In competitive markets, **demand = willingness to pay (WTP)** for private goods.
- For public goods, **individual WTP understates marginal benefit (MB)** because people can consume **even if they pay less (or nothing)**.
- ***Result:* Free riding** → (Total Voluntary Funding) < (Cost) → **Underprovision.**

Note

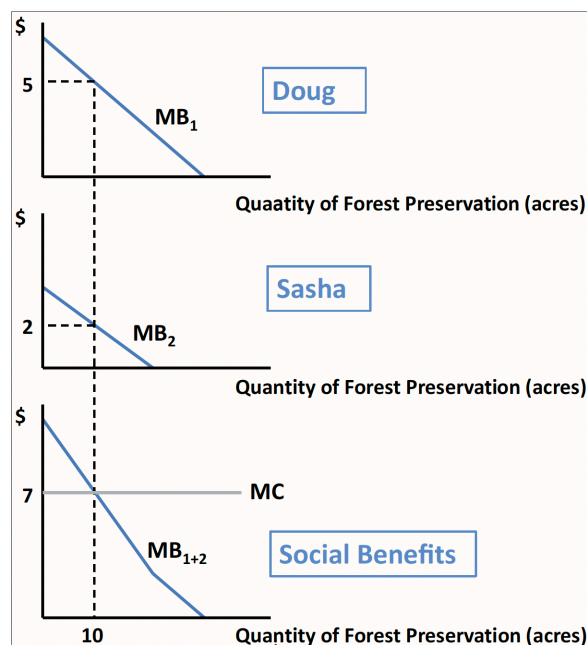
- Market failure here is **on the demand side**, unlike overuse of common-pool resources which is **on the production side**.

Aggregating Benefits: Vertical Summation



- Both value forest preservation (acres Q).
- At $Q = 10$, Doug's $MB = \$5$; Sasha's $MB = \$2$.
- For public goods, **social marginal benefit (SMB)** is the **vertical sum** of individual MB curves
 - **SMB = \$7.**
- Suppose **SMC = \$7 per acre** (constant).
- Then $Q^* = 10$ where **SMB = SMC**.

Funding the Optimum (Who Pays?)



- **Total Cost = \$70. Options:**
 - **Uniform tax:** Doug \$35, Sasha \$35 (simple, but ignores benefit differences).
 - **Benefit-based shares** (ideal but hard to measure): Doug pays more since his MB is higher.
 - **Ability-to-pay:** progressive finance to address equity concerns.
- Tensions:
 - **Efficiency:** hit Q^* .
 - **Fairness:** how to split the bill?

Policy Tools - Direct Public Provision

- Government directly **provides the public good**, funded through **general taxation**.
- Avoids reliance on voluntary contributions → ensures adequate and stable supply.
- Citizens share costs through taxes, but **everyone benefits** once the good is provided.
- *Examples:*
 - National defense
 - Basic scientific research
 - Public health campaigns (e.g., vaccination drives)

Policy Tools - Matching grants & subsidies

- Amplify voluntary contributions by adding public funds
Encourages private giving, lowers cost of participation
- *Examples:*
 - U.S. federal **charitable gift matching** programs (e.g., matching donations to conservation NGOs)
 - **Public broadcasting** pledge drives with matching challenges
 - **State and federal subsidies** for rooftop solar panels and energy-efficient appliances
 - Local governments matching **community fundraising** for parks, libraries, or schools
 - International funds (e.g., **Green Climate Fund**) co-financing climate adaptation projects with national contributions

Policy Tools - Earmarked Taxes & Levies

- **Dedicated funding streams** linked directly to a specific public good.
- Revenues cannot be spent elsewhere → builds **trust** that taxes support the intended purpose.
- Helps align **costs with beneficiaries** (those who pay are often those who benefit most).
- *Examples:*
 - Resilience bonds to finance disaster protection infrastructure
 - Carbon levies to support climate mitigation and clean energy
 - Local conservation assessments for parks, wetlands, or open space protection

Policy Tools - Patents & Prizes

- Provide **incentives for innovation** by making knowledge partly excludable.
- **Patents**: grant temporary monopoly rights to inventors → they can earn profits, encouraging investment in R&D.
- **Prizes**: reward specific breakthroughs without ongoing monopoly, while allowing broader diffusion.
- Both approaches aim to stimulate innovation while still creating **knowledge spillovers** that benefit society.
- *Examples*:
 - Pharmaceutical patents for new drugs
 - XPRIZE competitions (e.g., carbon removal, space technology)
 - Government innovation prizes for energy efficiency or medical devices

Policy Tools - Information & Social Norms

- Use **campaigns, education, and messaging** to influence behavior and public attitudes.
- Can shift **social norms** so that contributing to or supporting a public good becomes the default.
- Works best when combined with other policy tools (taxes, subsidies, regulation).
- *Examples*
 - Climate awareness campaigns (e.g., “Act on Climate”)
 - Energy-saving initiatives (“Turn off the lights,” Energy Star labeling)
 - Public health messages (seatbelt use, anti-smoking, vaccination reminders)
 - Recycling and waste reduction campaigns

Money Growing on Trees: Payments for Environmental Services in Community Forestry

Climate Change and Forests

- Climate change arises from **net greenhouse gas emissions**
- Abatement (mitigation) can come from
 - ↓ **Reducing sources** — burning fewer fossil fuels
 - ↑ **Increasing sinks** — forests that absorb CO₂
- Forests act as **natural carbon storage**
- “Avoiding deforestation” = *negative emissions relative to baseline*

Avoiding Deforestation: Economics of Abatement

- **Social benefit:** avoided climate damages (global public good)
- **Social cost:** opportunity cost of land or resources
 - Lost agricultural income
 - Lost timber or charcoal value
- Many tropical regions offer **low-cost abatement** opportunities
→ Why global climate finance often targets developing countries

REDD+ and PES

- Avoiding deforestation creates **positive externalities**
- **Payments for Ecosystem Services (PES):**
 - Pay local households for the environmental services they provide
 - Internalize the externality through incentive-compatible contracts
- Example: **REDD+**
 - *Reducing Emissions from Deforestation and Forest Degradation*
 - + Sustainable forest management for carbon storage

Who Pays for PES Contracts?

- Polluters (firms or countries) are often required by regulation to **limit their total emissions**.
- Instead of cutting emissions directly, they can **buy offsets**, such as:
 - “Protect 1 hectare of rainforest instead of emitting 100 tons of CO₂.”
- These offsets are integrated into **cap-and-trade** markets.
 - Companies can choose between **permits** and **offsets** to meet emission limits.
- International PES schemes therefore act as **global offsets**, transferring resources from **high-emission (often rich)** countries to **low-income** forest communities that conserve carbon sinks.

Why Is PES Hard?

Verifiability & Additionality

- **Verifiability** → Can we confirm the forest was truly protected?
 - Avoids *fraud* and *double counting*
- **Additionality** → Would it have been protected *anyway*?
 - Avoids *non-additional* payments
- Non-verifiable or non-additional contracts
 - Waste resources and fail to reduce total emissions

PES Role-Playing Game

You are a **household** in rural *Ephoria*, a low-income tropical country.

View the Game

- Forest resources = **subsistence + cash**
 - Government introduces a **PES contract**:
 - Pay \$50 if you agree not to harvest wood this year
 - Each round = new policy design
 - **Option A: Deforest**
 - Harvest wood for cooking or sale
- vs.**
- **Option B: Conserve**
 - Accept PES payment
 - Commit *not* to harvest

Contract Periods (Treatments)

Period	Policy Context	Key Feature
0	Baseline	No PES (individual harvest only)
1	PES	Legal PES contracts only
2	PES + Illegal Harvest	Can cheat; subject to random audit
3	Community PES	Group decision and shared payment
4	Community + Illegal Harvest	Group and individual enforcement

CP1 – *Baseline vs. PES*

- How did you decide between harvesting and PES?
- Did your **card value** affect your choice?
-  **Concepts**
 - Opportunity cost (abatement cost)
 - *Additionality* (True *additional* conservation only occurs when the payment **changes** behavior)
 - Heterogeneity in cost → cost-effective targeting

CP2 – *Illegal Harvest Option*

- You could sign PES but still harvest (“cheat”)
- Small chance of being **audited**
- If caught → lose payment + fine
-  **Key Idea:** Rational Theory of Crime (Becker 1968)
 - Expected benefit of cheating =
(gain if not caught) – (probability × penalty)
-  What share of the class cheated? Why?

CP3 – *Community Contract*

- Now, each **community decides together** whether to join PES
- If accepted, PES payment must be **shared** among members
- Split rules vary: equal, proportional, needs-based, etc.
- 🧡 **Concepts**
 - Community governance (Ostrom)
 - Fairness & equity norms
 - Coordination vs. free-riding

CP4 – *Community Contract + Illegal Harvest*

- Communities can **join PES** *and* decide whether to **police**
 - If policing → cost \$5 per household, no cheating possible
 - If not → some may harvest illegally
 - Audit probability increases with # illegal harvesters
-  **Concepts**
 - Self-governance
 - Collective enforcement
 - Trust & social capital
 - Deterrence by monitoring

The Promise of Forest-Based PES

- **Aligns incentives:** Encourages conservation by paying for the ecosystem services forests provide (carbon storage, biodiversity, water regulation).
- **Cost-effective mitigation:** Taps into *low-cost abatement opportunities* in developing regions — “climate finance with co-benefits.”
- **Income diversification:** Provides a stable cash flow to rural households that might otherwise rely on unsustainable logging or agriculture.
- **Global–local partnership:** Transfers resources from high-emission (often rich) countries to low-income communities maintaining forests.
- **Flexibility:** Can be scaled or customized — individual, community, or landscape-level programs.

⚠️ Potential Pitfalls and Challenges

- **Additionality problem**
 - CP1 in our game shows how payments can reward *non-additional* conservation.
- **Verification and enforcement**
 - Illegal harvesting in CP2 and CP4 highlights the *verifiability challenge*.
- **Leakage**
 - Conservation in one area can shift deforestation to another.
 - Not captured in our classroom game, but critical in real programs.
- **Equity and fairness**
 - Who gets paid — landowners or landless users?
 - How are community payments shared (as in CP3–CP4)?
 - Risk of elite capture or exclusion of marginalized groups.
- **Dependence on external funding**
 - Long-term sustainability uncertain once donor support ends.



Equity, Empowerment, and Governance

- PES can **empower communities** by recognizing their role in environmental stewardship — especially under community-based contracts (like CP3–CP4).
- However, unequal land rights and bargaining power can **reinforce inequalities** — wealthier households or leaders may capture the majority of benefits.
- Trust and local governance capacity determine whether PES fosters cooperation or conflict.
- The design of **payment division rules** (equal vs. proportional vs. needs-based) has real implications for fairness and motivation.

Reflective Prompts for Class Discussion

- In our experiment, who gained most from PES — low-card or high-card players?
- Did community-level decision-making improve or reduce fairness?
- If you were a policymaker in *Ephoria*, how would you redesign PES to:
 - Improve **additionality**,
 - Strengthen **compliance**, and
 - Ensure **equitable benefit sharing**?
-  **Key takeaway:**
 - PES succeeds not just when payments exist — but when they are *credible, additional, verifiable, and fair*.